Pesticide Biochemistry and Physiology 204 (2024) 106103

Contents lists available at ScienceDirect

Pesticide Biochemistry and Physiology

journal homepage: www.elsevier.com/locate/pest

# The biological activity of bacterial rhamnolipids on *Arabidopsis thaliana* and the cyst nematode *Heterodera schachtii* is linked to their molecular structure

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ARTICLE INFO

*Keywords:*
Rhamnolipids
*Heterodera schachtii*
*Arabidopsis thaliana*
Structure-activity relation
Plant priming
Reactive oxygen species

ABSTRACT

Rhamnolipids (RLs) are amphiphilic compounds of bacterial origin that offer a broad range of potential applications as biosurfactants in industry and agriculture. They are reported to be active against different plant pests and pathogens and thus are considered promising candidates for nature-derived plant protection agents. However, as these glycolipids are structurally diverse, little is known about their exact mode of action and, in particular, the relation between molecular structure and biological activity against plant pests and pathogens.

Engineering the synthesis pathway in recombinant *Pseudomonas putida* strains in combination with advanced HPLC techniques allowed us to separately analyze the activities of mixtures of pure mono-RLs (mRLs) and of pure di-RL (dRLs), as well as the activity of single congeners. In a model system with the plant *Arabidopsis thaliana* and the plant-parasitic nematode (PPN) *Heterodera schachtii* we demonstrate that RLs can significantly reduce infection, whereas their impact on the host plant varied depending on their molecular structure. While mRLs reduced plant growth even at a low concentration, dRLs showed a neutral to beneficial impact on plant development. Treating plants with dRLs triggered an increased reactive oxygen species (ROS) production, indicating the activation of stress-response signaling and possibly plant defense. Pretreatment of plants with mRLs or dRLs prior to application of flagellin (flg22), a known ROS inducer, further increased the ROS response to flg22. While dRLs stimulated an elevated flg22-induced ROS peak, a pretreatment with mRLs resulted in a prolonged synthesis of ROS indicating a generally elevated stress level. Neither mRLs nor dRLs induced the expression of plant defense marker genes of salicylic acid, jasmonic acid, and ethylene pathways.

Detailed studies on dRLs revealed that even high concentrations up to 755 ppm of these molecules have no lethal impact on *H. schachtii* infective juveniles. Infection assays with individual dRL congeners showed that the C10-C8 acyl chained dRL was the only congener without effect, while dRLs with C10-C12 and C10-C12:1 acyl chains were most efficient in reducing nematode infection even at concentrations below 2 ppm. As determined by phenotyping and ROS measurements, *A. thaliana* reacted more sensitive to long-chained dRLs in a concentration-dependent manner.

Our experiments show a clear structure-activity relation for the effect of RLs on plants. In conclusion, functional assessment and analysis of the mode of action of RLs in plants and other organisms require careful consideration of their molecular structure and composition.

*Abbreviations:* RLs, Rhamnolipids; dRLs, di-rhamnolipids; mRLs, mono-rhamnolipids; CI, chemotaxis index; ddH2O, double distilled water; DPI, days post inoculation; DPS, days post sowing; flg22, flagellin 22; HAAs, 3-(3- hydroxyalkanoyloxy)alkanoic acids; HRP, horseradish peroxidase; J2, Heterodera schachtii second stage juvenile; MAMP, microbe-associated molecular pattern; NAMP, nematode-associated molecular pattern; PAMP, pathogen-associated molecular pattern; ppm, parts per million; RLU, relative light units; ROS, reactive oxygen species.

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https://doi.org/10.1016/j.pestbp.2024.106103
Received 4 June 2024; Received in revised form 18 August 2024; Accepted 24 August 2024
Available online 26 August 2024
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## 1. Introduction

Rhamnolipids (RLs) are glycolipidic biosurfactants produced as secondary metabolites by different bacteria, of which *Pseudomonas* spp. and *Burkholderia* spp. are studied best. The synthesis is regulated by quorum sensing in a cell-density-dependent manner and synchronizes with the stationary phase of bacterial growth (Sodagari and Ju, 2020). A set of three genes, namely *rhlA*, *rhlB* and *rhlC*, is responsible for the final synthesis steps from the fatty acid precursors to 3-(3- hydroxyalkanoyloxy)alkanoic acids (HAAs) to mono-RLs (mRLs) with one rhamnose molecule and eventually to di-RLs (dRLs) with two rhamnose molecules as hydrophilic the head group (Wittgens et al., 2017). The lipophilic tail plays a pivotal role in the structural diversity of these molecules via variations in chain length, degree of saturation, and the number of fatty acid chains attached to the head group. Until today, there are about 110 congeners described and the congener range and composition depend on the producer strain and external stimuli such as nutritional parameters (Blunt et al., 2022; Chen et al., 2017; Guzmán et al., 2024; Reis et al., 2011; Rikalović et al., 2015). Hence, variation in RL synthesis also mirrors environmental requirements and challenges and the specific RL composition shapes its character as a specific adjustment to environmental conditions. Tensioactive properties like the critical micelle concentration (CMC) range from 5 to 200 ppm due to the differences in the ratio of mRLs to dRLs (Rikalovic et al., 2013). Information on the influence of the size of the head group on micelle formation is inconsistent though (Rikalovic et al., 2013; Zhou et al., 2019), as the production, purification, and analytics is not straight forward. In general, it is assumed that the competency of a RL mixture mainly depends on the properties of the most abundant congener (Rikalović et al., 2015). The biological role of RLs is usually associated with their tensioactive properties contributing to nutrient availability, biofilm formation, and bacterial motility, but it also involves quorum sensing and antimicrobial activity (Chrzanowski et al., 2012; Soberón-Chávez et al., 2005). In particular, the numerous reports of the antimicrobial activity of RLs indicate an outstanding potential of these bacterial molecules for example as biologicals in environmentally friendly crop protection (Chen et al., 2017; Zhao et al., 2022). The question regarding their mode of action remains to be answered and the answer appears to combine multiple aspects. Anti-oomycete activity of RLs is often associated with direct lysis of zoospores due to the absence of a thick cell wall while antibacterial activity is less radical and proposed to depend on cell permeabilization and changes in cell hydrophobicity (Stanghellini and Miller, 1997; Vasileva-Tonkova et al., 2011). Several studies describe plant immune stimulation by RLs (Robineau et al., 2020; Sanchez et al., 2012; Varnier et al., 2009), but the individual steps starting with the nature of RL perception to early signaling processes are still a subject of speculation. Schellenberger et al. (2021) observed an alteration in early signaling in the double mutant AtFAH1/2 (*Fatty Acid Hydroxylase 1/2*) upon RL treatment. The mutations result in changes in the sphingolipid composition and a general disorder of the plasma membrane of Arabidopsis. Hence, membrane integrity must play a role in RL recognition.

A key obstacle in understanding the mode of action is the lack of comparability between studies due to insufficient definition of differences in RL compositions. Despite the well-described differences in the biophysical character of individual RL types and congeners, literature aiming to identify the relation between structure and biological activity is still relatively sparse in comparison to everything that has already been published about the efficacy of RLs, particularly in the context of plant-parasitic nematodes (PPNs). Therefore, we investigated the efficacy of different RLs to impair the host-parasite interaction in a model system with *Arabidopsis thaliana* and the PPN *Heterodera schachtii*. As a cyst nematode, this species is among the most economically important PPNs, and is also an excellent model for studying parasitic interactions of PPNs with plants (Jones et al., 2013; Sijmons et al., 1991) as well as the effect of different molecules and organisms (Huang et al., 2021; Matera et al., 2021; Schleker et al., 2022). Effective RL mixtures were further analyzed with focus on their influence on plant immunity including the expression of defense markers and their roles as direct and indirect elicitors of oxidative signaling. Eventually, the effect of the most efficient RL type was evaluated on the level of RL congeners.

## 2. Material and methods

### 2.1. Biosurfactant production

HAAs, mRLs, and dRLs were produced with three different recombinant *Pseudomonas putida* KT2440 strains using the *rhl* gene set originating from *Pseudomonas aeruginosa*. *P. putida* KT2440 pSB01 (Tiso et al., 2017) was used for the production of HAAs, while *P. putida* KT2440 SK4 was used for the production of mRLs (Tiso et al., 2020a), and *P. putida* KT2440 pWJ02 was used for the production of predominantly dRLs (Tiso et al., 2017). The respective HAAs and RLs were a mixture of congeners with variations in the chain length of the β-OH fatty acids.

The strains were routinely grown on Luria Bertani (LB) agar plates. Cultivation was performed in Fernbach flasks filled with 500 ml LB medium supplemented with 20 g/l glucose and incubated in a Minitron Shaking Incubator (Infors AG, Bottmingen, Switzerland) at 30 °C, 200 rpm, and 50 mm shaking diameter. LB medium contained 10 g/l tryptone, 5 g/l yeast extract, and 5 g/l NaCl at pH 7.2. LB agar plates additionally contained 8 g/l agar. The final medium was sterilized by autoclaving for 20 min at 121 °C before adding sterile glucose. For cultivation of *P. putida* KT2440 pSB01 and *P. putida* KT2440 pJW02, kanamycin with a concentration of 50 μg/ml was added to maintain selection pressure. Overnight cultures were used to inoculate main cultures by adjusting the optical density at 600 nm to 0.1.

### 2.2. Biosurfactant quantification

For quantification of hydroxyfatty acids, HAAs and RLs, components of the samples were separated using reversed-phase chromatography as described by Tiso et al. (Tiso et al., 2016), based on the method established by Behrens et al. (Behrens et al., 2016). As HPLC system, a DIONEX UltiMate 3000 (Thermo Fisher Scientific, Inc., Waltham, MA, USA) composed of an LPG-3400SD pump, a WPS-3000 (RS) autosampler, a TCC-3000 (RS) column oven, and a Corona charged aerosol detector (CAD) was used. The detector was supplied with a continuous nitrogen stream by a Parker Balston NitroVap-1LV nitrogen generator (Parker Hannifin GmbH, Kaarst, Germany). A NUCLEODUR C18 Gravity column from Macherey-Nagel GmbH and Co. KG (Düren, Germany) with a precolumn cartridge of 4 mm length, a particle size of 3 μm, and dimensions of 150 × 4.6 mm was utilized to separate the biosurfactants. The gradient program started with 70 % (*v*/v) acetonitrile and 30 % (v/v) ultrapure water containing 0.2 % (v/v) formic acid. The acetonitrile percentage was constantly increased until it reached 80 % (v/v) at 8 min and then kept constant for 1 min before it was further increased to 100 % (v/v) within 1 min. After 11 min of total running time, the acetonitrile percentage was decreased again back to 70 % (v/v) within 1.5 min to retrieve starting conditions. The total analysis time was 15 min. The column oven temperature was set to 40 °C and 5 μl of the sample were injected. The flow rate was set to 1 ml/min. Samples were prepared by centrifugation at 13,000 ×*g* for 1 min, whereupon 400 μl of the cell-free supernatant were mixed with 400 μl acetonitrile, and the resulting mixture was vortexed. Subsequently, samples were incubated overnight at 4 °C to facilitate precipitation of proteins and other residuals that might clog the column. Next, biomass and precipitated material were removed by centrifugation at 13,000 ×*g* for 1 min. Before HPLC analysis, the supernatant was filtered with a Phenex-RC (regenerated cellulose) syringe filter (diameter, 4 mm; pore size, 0.2 μm; Phenomenex, Torrance, CA, USA).

### 2.3. Biosurfactant purification

Purification of HAAs and RLs was based on a method described previously (Tiso et al., 2020b). Prior to purification, cells and proteins were removed from the culture broth by addition of acetonitrile as specified above. Isolation of the biosurfactants from the supernatant was then performed by solid phase extraction (SPE) using a C18 derivatized silica-based adsorbent (AA12SA5, YMC Europe GmbH, Dinslaken, Germany) packed into a Bioline glass column (Knauer GmbH, Berlin, Germany). For desorption, an AZURA analytical pump P 6.1 l (Knauer GmbH, Berlin, Germany) was employed to carry out a gradient elution, starting with 100 % ultrapure water and ending with 100 % pure ethanol. The flow rate was set to 1 ml/min and a total of 21 fractions of 10 ml each were collected using a 2111 Multirac fraction collector (LKB, Bromma, Sweden). Fractions containing the desired RL congeners were combined and the solvent was evaporated.

For subsequent chromatographic separation of single RL congeners, the dried products were resolved in a 50/50 (*v*/v) acetonitrile/water mixture. The separation was carried out by the use of a preparative HPLC system consisting of an AZURA analytical pump P 6.1 l, an AZURA autosampler 3950 (both Knauer GmbH, Berlin, Germany), a SEDEX 58 LT-ELSD detector (SEDERE, Olivet, France), and a Foxy R1 fraction collector (Teledyne ISCO Inc., Lincoln, NE, USA). As centerpiece of this setup, a NUCLEODUR C18 HTec column from Macherey-Nagel GmbH and Co. KG (Düren, Germany) with a precolumn cartridge of 16 mm length, a particle size of 5 μm, and dimensions of 250 × 21 mm was employed. The flow rate was set to 10 ml/min and a total amount of 200 mg of HAAs, 100 mg of mRLs or 10 mg of dRLs were injected. Elution was carried out starting with 70 % (*v*/v) acetonitrile and 30 % (v/v) 0.2 % (v/v) formic acid. For the separation of HAAs and mRLs, acetonitrile was constantly increased until it reached 85 % (v/v) at 45 min and then further increased to 100 % (v/v) within 5 min. After 56 min of total running time, the acetonitrile percentage was decreased back to 70 % (v/v) within 2 min to retrieve starting conditions. The total running time was 60 min. For the separation of dRLs, the acetonitrile percentage was kept constant for 55 min and then increased to 100 % (v/v) within 3 min. After 61 min of total running time, the acetonitrile percentage was decreased back to 70 % (v/v) within 4 min to retrieve starting conditions. The total running time was 70 min. The desired RL congeners were fractionated according to peak elution and subsequently evaporated to obtain pure, solvent-free biosurfactants. The dried products were solved in ultrapure water and the pH was adjusted to 7.0 by the use of 0.1 M NaOH and 0.1 M HCl. Finally, the preparations were filtered with a ROTILABO RC syringe filter (diameter 25 mm; pore size 0.2 μm; Carl Roth GmbH & Co. Kg, Karlsruhe, Germany) to guarantee sterility.

The final HAA mixture contained 18 % C10-C8, 50 % C10-C10, 18 % C10-C12, and 1 % C10-C12:1. The final mRL mixture contained 10 % C10-C8 mRLs, 62 % C10-C10 mRL, 17 % C10-C12 mRLs, and 12 % C10-C12:1 mRLs. The final dRL was comprised of 5 % C10-C8 dRLs, 61 % C10-C10 dRLs, 19 % C10-C12 dRLs, and 14 % C10-C12:1 dRLs (Suppl. Fig. 1). In all our experiments ppm corresponds to mg/l. Due to the high stock concentrations, RLs and HAAs were thoroughly vortexed for one minute to circumvent micelle formation right before use.

### 2.4. Phenotyping of host-pathogen interaction

Seeds of *A. thaliana* wild type Col-0 were sterilized using 70 % ethanol for 30 s, followed by 1.3 % sodium hypochlorite for three minutes. Sterile seeds were immediately washed three times in sterile double distilled water and dried onto sterile filter paper. For stratification, seeds were stored at 4 °C until further use. Sowing was performed with five seeds per Petri dish filled with Knop agar (Sijmons et al., 1991) and stored at room temperature with 16/8 photoperiod in a tilted position to promote root surface growth for a facilitated transfer eight days post sowing (DPS). For the transfer, the plants were separated and two well- and similarly developed plants were transferred onto a new plate filled with about 16 ml Knop agar supplemented with the RL or HAA as respectively mentioned or standard Knop agar as control. After two days of adaption, per plant 60–70 freshly hatched *H. schachtii* J2s (hereafter J2s) (Matera et al., 2021) were inoculated in two drops along each primary root. Infection evaluation started 12 days post inoculation (DPI) by counting female and male nematodes under a Stereo Microscope (Leica, Germany), followed by female size measurement 14 and 28 DPI using the LAS software (Leica Microsystems), and eventually female collection 35 DPI to invasively count the eggs and juveniles. The plant phenotype was ascertained 14 DPI by first separating shoot and root tissue at the hypocotyl and immediately weighing the fresh weight (FW) of the shoot with a fine balance. In order to remove the complete root system without breakage the agar was first dissolved in a microwave and the agar-free root system was transferred into a transparent tray designed for subsequent root scanning (Epson Perfection V700 Photo root scanner) connected to the WinRHIZO Pro software (Regent Instruments, Québec, Canada) for automatic computation of the total root length.

### 2.5. Nematode mortality assay

A transparent flat bottom 96-well cell culture plate was prepared by filling a minimum of four wells per variant and time point with 100 μl dRLs (188.75, 377.5, 755, and 1510 ppm) or double distilled water as control further containing about 50 J2s prepared according to Matera et al. (2021). Incubation was performed at room temperature and mortality was assessed every 24 h for four days using 10 μl of 5 M sodium hydroxide (NaOH) per well with four technical replicates per time point and treatment. NaOH provokes movement of live nematodes and thus allows reliable differentiation between alive and dead nematodes (Matera et al., 2021).

### 2.6. Nematode activity assay

Standard or Knop agar supplemented with mRLs or dRLs (0.25, 2.5, or 25 ppm) in 100 mm petri dishes was inoculated with approximately 50 freshly hatched J2s, prepared following Matera et al. (2021), at the plate center. The inoculation drop was allowed to air dry in the clean bench before the plates were closed. After incubation at RT for 4 h, the number of nematodes located beyond a distance of 4.5 cm from the plate center were counted and set into relation to the total number of nematodes in the same plate to assess nematode activity by percentage.

### 2.7. Nematode chemotaxis assay

Chemotaxis activity of RLs towards *H. schachtii* was assessed using a modified protocol by Choe et al. (2012). Standard Knop agar petri dishes (100 mm) were prepared with 10 μl of either 1 μM, 1 nM, and 1 pM mRL-C10-C10 or dRL-C10-C10 placed 3.75 cm from the plate center, 10 μl double distilled water was placed in the same distance to the center opposite to the test agent. Approximately 100 J2s were inoculated onto the center, prepared following the protocol by Matera et al. (2021). The inoculation drop was allowed to air dry in the clean bench before the plates were closed. After incubation at RT for 4 h, the number of nematodes that were located within a radius of 2.5 cm of each spot were counted. The chemotaxis index (CI) was determined by subtracting the number of J2s in the control from the number of J2s in the RL spot area divided by the total number of J2s in the same plate (Bargmann et al., 1993; Margie et al., 2013).

### 2.8. Reactive oxygen species analysis

Reactive oxygen species (ROS) biosynthesis was assessed using a modified protocol following Melcher and Moerschbacher (2016) and Mendy et al. (2017). *A. thaliana* was cultivated on standard Knop agar as described above, but with four seeds per plate. The Petri dishes were

stored in a tilted position to ensure surface growth of the roots to facilitate damage-free root removal. About two weeks old leaves were harvested by cutting at the leaf base ensuring a similar size between all plant leaves and optimal fit into the well of a 96-well plate while the whole root system was cut at the hypocotyl.

For the analysis of RLs as direct trigger of ROS, the root system was incubated in sterile double distilled water overnight in the dark the evening prior to the measurement to diminish the initial plant stress level. The following morning, a white flat bottom 96-well plate was filled with 80 μl of 100 μM L-012 sodium salt (©FUJIFILM Wako Pure Chemical Corporation) per well. Roots were gently tapped dry and carefully transferred making sure that all root tissue was in the liquid. Prior to the measurement each well was further filled with 20 μl of horseradish peroxidase (HRP, 20 μg/ml) and 50 μl of either the test solution as threefold concentration or sterile double distilled water as negative control and 1 μM of the flg22 as positive control. The 96-well plate was instantly loaded into a microplate reader (Infinite ® 200 PRO by Tecan Life Sciences Home, Germany) and luminescence was measured as relative light unit in a two-minute interval over a period of two hours.

For the analysis of the effect of a pretreatment with RLs, leaf tissue was incubated in mRLs or dRLs (control with sterile double distilled water) (Table 1) overnight in the dark the evening prior to the measurement. The next morning, leaves were carefully transferred using a fine brush. The subsequent steps were performed as described above with the exception that ROS was triggered with flg22 only. Sterile double distilled water was used as negative control.

### 2.9. Gene expression analysis

*A. thaliana* wild type Col-0 seeds were sterilized and cultivated on standard Knop agar as described above, but with four seeds per plate. After eight days well-developed plants were transferred onto Knop agar supplemented with either 8.3 ppm mRLs or dRLs. The root system was sampled 1 h and 48 h after transplanting and transferred into a 1.5 ml Eppendorf tube for immediate freezing in liquid nitrogen and subsequent storage at −80 °C until further use. Root total RNA was extracted with the RNeasy Plant Mini Kit (QIAGEN, Germany), genomic DNA contamination was removed with TURBO DNA-free™ Kit (Invitrogen, Germany) und reverse transcription was performed with the High-Capacity cDNA Reverse Transcription Kit (Thermofisher, Germany) and 100 ng RNA per reaction (Thermofisher) according to the respective manufacturer's instructions. RT-PCR implied 20 μl per reaction with 10 μl 2× SYBR™ Green PCR Master Mix (Thermofisher), 0.5 μl of forward and reverse primer each (10 μM) and 1 μl cDNA (diluted 1:5) filled up with molecular grade water. The primer sequences can be found in the supplementary table 1 and, apart from MYB72, were all previously published (Kammerhofer et al., 2015; Pinfield-Wells et al., 2005; Sanchez et al., 2012). Expression values were normalized to the expression of UBIQUITIN-SPECIFIC PROTEASE 22 (UBP22) (Hofmann and Grundler, 2007). One root system represented one biological replicate and was measured as three technical replicates.

**Table 1**
Overview of the variants in the plant leaf ROS assay to analyze the effect of a pretreatment with mono-rhamnolipids (mRLs), di-rhamnolipids (dRLs) or double distilled water ($ddH_2O$) as control prior to eliciting plant ROS response with flagellin 22 (flg22).

| Variant | Pretreatment | Trigger |
|---|---|---|
| water/water | $ddH_2O$ | $ddH_2O$ |
| water/flg22 | $ddH_2O$ | flg22 |
| mRLs/water | mRLs | $ddH_2O$ |
| mRLs/flg22 | mRLs | flg22 |
| dRLs/water | dRLs | $ddH_2O$ |
| dRLs/flg22 | dRLs | flg22 |

### 2.10. Statistical analysis

Statistical analysis and graphs were generated with SigmaPlot (Version 12.5, Systat Software GmbH; Germany). For each experiment, the data of up to three independent biological replicates in total were analyzed. Normality was tested based on Shapiro-Wilk with a *P*-value of 0.05 for rejection. Significance of multiple comparison analyses was calculated based on ANOVA on ranks and an appropriate post-hoc test. Tukey test was the post-hoc analysis of choice for experiments with an equal number of technical replicates, and when there were five or more groups to compare. Student Newman-Keuls (SNK) method was used for experiments with an equal number of technical replicates, and when there were less than five groups to compare whereas the Dunn's method was used for an unequal number of technical replicates among variants. Fold changes in root gene expression were normalized to the expression of the control. Significance in root gene expression was analyzed based on an unpaired *t*-test in comparison to the control.

## 3. Results

### 3.1. Effect of RLs on nematode infection and plant growth

The effect of RLs on *H. schachtii* infection was first validated with focus on the amphiphilic structures of their final biosynthesis steps, namely of HAAs, mRLs or dRLs. For assessment, eight-day-old *A. thaliana* seedlings were transferred to compound-supplemented plant growth medium and inoculated with *H. schachtii* J2s 2 days later.

RLs clearly affected nematode infection at a concentration of only 8.3 ppm (parts per million, equals mg/l), whereas the same concentration of HAAs was ineffective (Fig. 1). Thus, it required the complete biosurfactant molecule comprising L-rhamnose and HAAs to significantly reduce the number of adult nematodes per plant (Fig. 1a), even though no clear adverse effects in female size development and nematode reproduction could be observed (Fig. 1b, c). In a concentration-dependent manner, but independent of the RL type, mRLs and dRLs both also affected the compulsion of J2s to move (Suppl. Fig. 2). The results of a chemotaxis assay performed with very low concentrations of the predominant C10-C10 congener of mRLs and dRLs indicate that mRLs, in contrast to dRLs, may have a subtle repellent effect on J2s of *H. schachtii* (Suppl. Fig. 3). Structure-related effects of RLs are even more distinct regarding host plant development. An HAA concentration of only 8.3 ppm reduced root growth by about 15 % whereas the same concentration of mRLs seemed to stop plant growth of above and below ground tissue (Fig. 1d, e, Suppl. Fig. 4). However, although the plants looked stressed showing a reddish discoloration of the leaves, no necrotic changes could be observed (not shown). In contrast, dRLs increased root growth by about 15 % and simultaneously reduced the infection by about 75 % (Fig. 1a, d). This experiment showed that HAAs, mRLs, and dRLs clearly have different effects on plant and nematode development.

As HAAs did not impact nematode infection, we then focused on analyzing the effects of mRLs and dRLs. We first determined the root expression of several defense-related genes upon direct exposure of roots to 8.3 ppm mRLs or dRLs for 1 h and 48 h. The later time point corresponds to the state of plant defense J2s encountered in the infection assays after inoculation. Neither a treatment with mRLs nor with dRLs resulted in considerable variation in the expression of selected marker genes compared to the control (Suppl. Fig. 5). Next, we investigated whether and how mRLs and dRLs affect plant ROS production upon a subsequent pathogen trigger simulated by the well-described ROS elicitor flg22 (3.2, Fig. 2, Suppl. Fig. 6). Changes in the ROS profile or total ROS production then indicate a stress response as well as changes in plant immunity induced by the RL treatment.

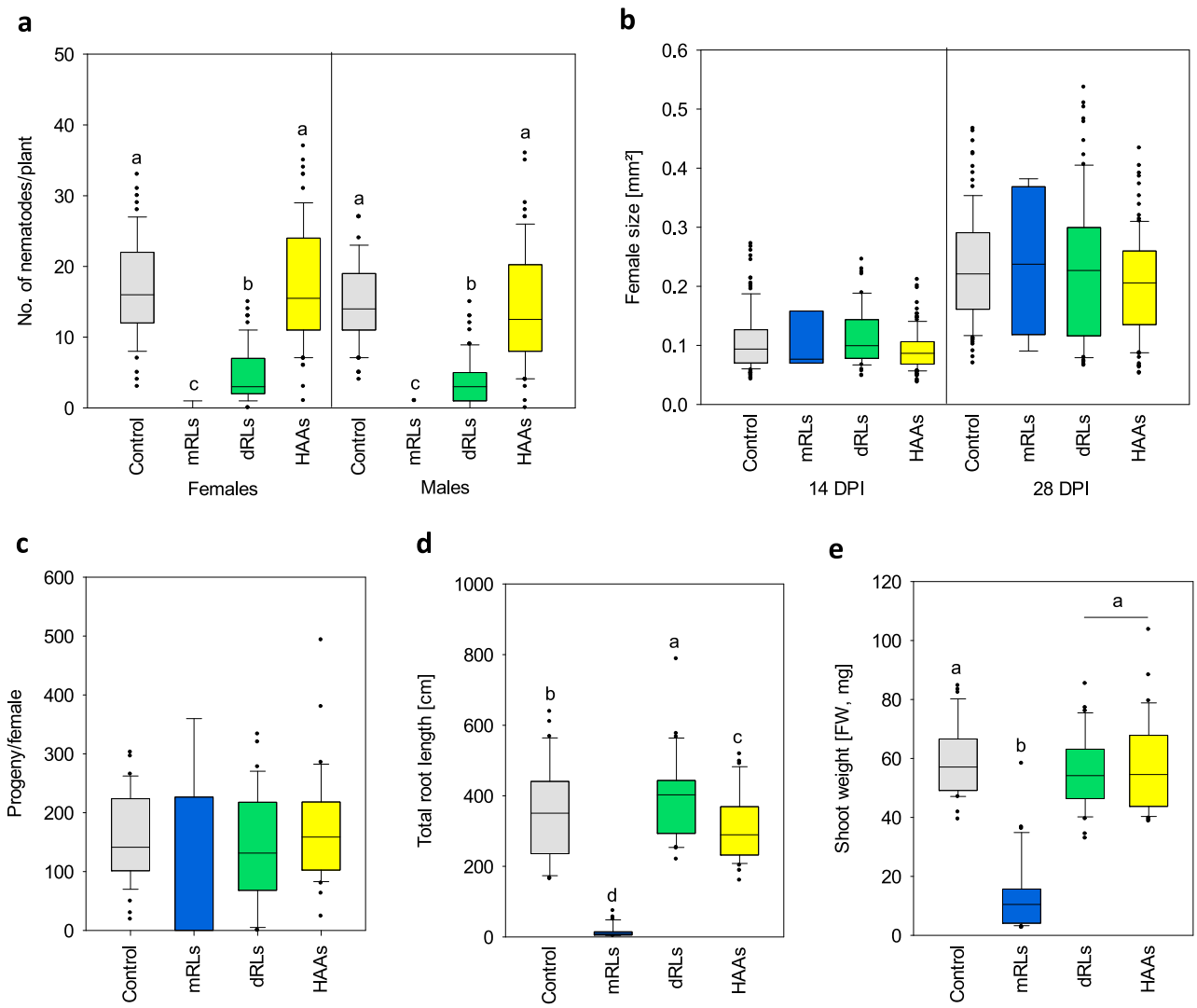


**Fig. 1.** Impact of different rhamnolipids (RLs) and 3-(3- hydroxyalkanoyloxy)alkanoic acids (HAAs) on nematode parasitism and the host plant *Arabidopsis thaliana* inoculated with *Heterodera schachtii* under in vitro conditions. The compounds were supplemented to the plant growth medium in a concentration of 8.3 ppm and their impact on (a) the number of adult nematodes at 14 days post inoculation (DPI), (b) the size of female nematodes at 14 and 28 DPI, and (c) progeny per female at 35 DPI, (d) total root length (cm) and (e) shoot fresh weight (mg) was analyzed at 14 DPI. Median with 25 %- and 75 %- quartile of three independent biological replicates with total $n = 70$ plants is shown for the nematode data and total $n = 34$ plants for the plant data. Significance is based on ANOVA on ranks and post-hoc Student–Newman–Keuls (SNK), different letters indicate significant differences between treatments ($p < 0.05$). Mono-RLs (mRLs), di-RLs (dRLs).

### 3.2. *Effect of mRLs and dRLs on ROS synthesis of A. thaliana induced by flg22*

Synthesis of extracellular ROS is among the earliest plant responses to certain environmental influences such as the stimulation by microbe-associated molecular patterns (MAMPs) (Marcec and Tanaka, 2022). The bacterial peptide flg22 is a well-studied MAMP known to induce a characteristic ROS profile in Arabidopsis leaves (Marcec and Tanaka, 2022) and thus is an excellent model system to detect alterations in plant immunity. Therefore, we tested whether a pretreatment with RLs modifies the ROS production in leaves in response to flg22 and thus primes plants prior to MAMP exposure (Fig. 2, Suppl. Fig. 6). Leaves of *A. thaliana* were harvested and incubated overnight in different concentrations of mRLs, dRLs or water as control. The following day, RLs were washed off and ROS levels in response to water or flg22 were measured over a period of two hours (Table 1).

Major differences between mRLs and dRLs could be observed. A pretreatment with 10 ppm mRLs led to a significant increase in total leaf ROS production either upon elicitation with flg22 or with water instead compared to the water/flg22 or the water/water variant, respectively (Fig. 2a). This indicates that mRLs already increase the basal ROS production. None of the dRL pretreatments caused a significant increase of total leaf ROS production (Fig. 2a). Interestingly, the only variant that significantly increased the peak height of the flg22-induced curve in comparison to the water/flg22 variant was the pretreatment with 10 ppm dRLs (Fig. 2c). A comparison of the time-resolved ROS profiles shows that leaf ROS production of the 10 ppm dRLs/flg22 variant reaches the level of the water/flg22 variant after about 50 min whereas the ROS level of the 10 ppm mRLs/flg22 variant remains more than twice as high compared the water/flg22 variant until the measurement was terminated at 120 min (Suppl. Fig. 6). None of the RL-pretreatments affected the time at which the peak tip was reached (Fig. 2b).

Both RL types altered the ROS profile for flg22 towards a stronger reaction. While a dRL pretreatment stimulated a temporary elevated ROS peak, a pretreatment with mRLs elevated the basal ROS level strongly indicating that Arabidopsis responds differently to both RL types and generally reacts more sensitive to mRLs. This could explain the observed Arabidopsis growth depression and hints to the possibility that the effect observed on plant susceptibility to *H. schachtii* resulted from impaired host plant qualities due to a decrease in root mass. In contrast,

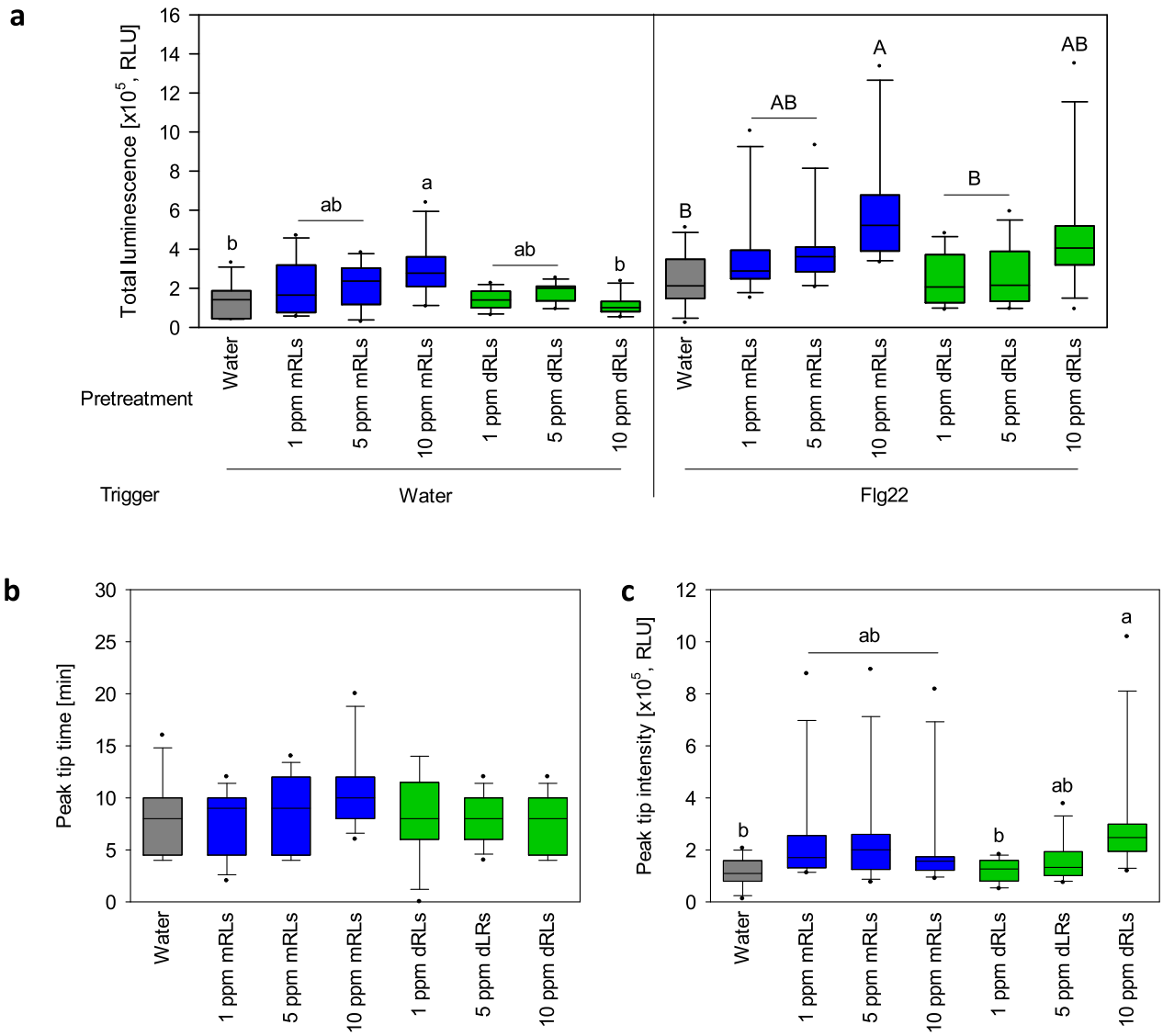


**Fig. 2.** ROS production of *Arabidopsis thaliana* leaf tissue triggered by flagellin 22 (flg22) after pretreatment with different concentrations of mono-rhamnolipids (mRLs, blue) or di-rhamnolipids (dRLs, green) monitored over a period of two hours. (a) Total ROS produced within the initial 2 h of flg22 induction, (b) time point at which the flg22-triggered curve of the assay shown in 2a reaches its peak after pretreatment with mRLs or dRLs, or double distilled water for control, (c) luminescence peak of the flg22-triggered curve of the assay shown in 2a after pretreatment with mRLs or dRLs, or double distilled water for control. Median with 25 %- and 75 %- quartile of three independent biological replicates with total $n = 11$ for water-triggered leaves and $n = 12$ for flg22-triggered leaves. Significance is based on ANOVA on ranks and post-hoc Tukey test, different letters indicate significant differences between treatments ($p < 0.05$). Relative Light Units (RLU). Corresponding ROS profiles are available in Suppl. Fig. 6. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

dRLs could reinforce plant immunity as observed by a stronger ROS response to the MAMP flg22 and reduced nematode infection without phenotypical disadvantages for the host plant. Thus, further experiments focused exclusively on dRLs.

### 3.3. Nematicidal effect of dRLs

A mortality assay with the nematode in its pre-parasitic stage (J2) was performed in order to understand whether a direct lethal impact of dRLs could be responsible for the observed decrease in infection. Exposure of J2s to dRL concentrations in the range used in the infection assay was ineffective (data not shown). An increase in concentrations to approximately 45 times as high as used in the infection assays did also not significantly increase nematode mortality (Fig. 3). Hence, it is unlikely that the decrease in the number of infected nematodes resulted from a direct nematicidal effect of dRLs.

### 3.4. Effect of single dRL congeners on nematode infection and plant

In a final step for understanding the structural relevance of RLs, the impact of single dRL congeners on nematode infection and plant ROS production was analyzed. In the infection assay, each dRL congener was studied in two different concentrations. The first concentration was chosen to compare the efficacy of all congeners at the same concentration (Fig. 4). The second concentration was chosen with reference to the hypothesis of the most dominant congener determining the overall character of the mixture (Fig. 5). Here, each congener was supplemented based on the proportion given in the dRL mixture used in the previous experiments at 8.3 ppm: 5 % C10-C8 dRL, 61 % C10-C10 dRL, 19 % C10-C12 dRL, and 14 % C10-C12:1 dRL.

None of the individual dRL congeners could entirely replicate the effect induced by the dRL mixture, but apart from C10-C8 dRLs all of them were able to interfere with nematode establishment in the root.

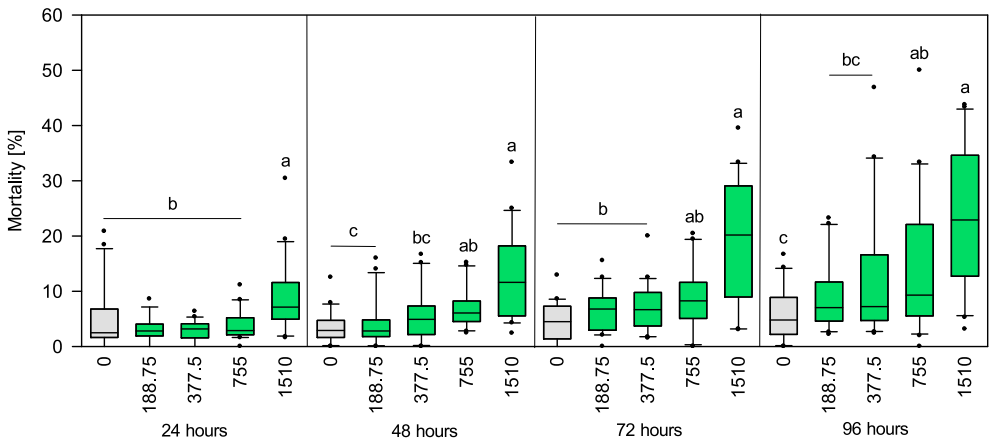


**Fig. 3.** Effect of di-rhamnolipids (dRLs) on the mortality of *Heterodera schachtii* infective stage juveniles (J2s). Nematodes were kept in solutions of dRLs at different concentrations (in ppm) for a period of four days. Median with 25 %- and 75 %-quartile of three independent biological replicates with in total $n = 20$ wells (about 50 J2s/well) per treatment and time point. Significance based on ANOVA on ranks and post-hoc Tukey test, different letters indicate significant differences between treatments ($p < 0.05$).

The efficacy was not only dependent on the respective concentration, but correlated with the 3-OH acyl chain length. RLs with longer chains appeared to be more efficient at lower concentrations. Very low concentrations of below 2 ppm of C10-C12 and C10-C12:1 dRLs reduced the infection as effective as the predominant congeners C10-C10 dRL with more than 5 ppm (Fig. 5a). Notably, root and shoot growth was significantly impaired during root exposure to 8.3 ppm of these longer dRL congeners (Fig. 4d, e).

We continued with investigations on root ROS synthesis triggered by 50 or 200 ppm single dRL congeners. In preliminary experiments, with RLs that slightly differed in the composition compared to the RLs used in this assay, ROS with lower concentrations did not show a clear ROS burst, thus we chose 50 and 200 ppm. A comparison of the individual root responses revealed plant sensitivity to longer dRL congeners (Fig. 6). A concentration of 50 ppm of C10-C12 and C10-C12:1 dRLs caused a significant increase in total ROS production compared to the water control. A concentration of 200 ppm did not induce ROS synthesis for none of the congeners or the dRL mixture. A comparison between the dRL mixture and its single congeners revealed no significant differences in the total ROS synthesis, but different patterns in the ROS profile (Suppl. Fig. 7). Exclusively upon exposure to C10-C12 and C10-C12:1 dRLs, there was a small peak at 50 ppm within less than five minutes that could be similarly observed for the 50 ppm dRL mixture. Another clear difference in the ROS profile is that root exposure to C10-C10 dRLs, the main congener of the mixture, displays the lowest ROS synthesis over the monitored time period.

In priming experiments, neither a pretreatment with the dRL mixture nor with the individual congeners induced a generally higher total leaf ROS synthesis (Fig. 7a), but affected the curve peak profile induced by flg22. Although there was no significant difference in the time it takes to reach the curve peak, there was a clear trend that the higher concentration of 10 ppm shortened the required time period in comparison to 1 ppm (Fig. 7b, Suppl. Fig. 8). Pretreatment with all dRLs variants led to a higher ROS level at the flg22-induced peak, thus a more intense reaction, but in comparison to the water/flg22 control only the value for the 10 ppm dRLs/flg22 variant was significantly increased (Fig. 7c).

Hence, the individual dRL congeners differed in their impact on nematode infection and on the plant both, in comparison to each other as well as in comparison to the complete mixture with a specific sensitivity towards longer dRL congeners.

## 4. Discussion

The structure-activity relation of RLs is still unclear when it comes to their ability to counteract plant pathogens like PPNs. Recently, it was found that certain HAAs are perceived by the bulb-type lectin receptor kinase LIPOOLIGOSACCHARIDE-SPECIFIC REDUCED ELICITATION/ S-DOMAIN-1-29 (LORE/SD1–29), which upon HAA induction stimulates local resistance in *A. thaliana* leaves to bacterial infection with *P. syringae* pathovar *tomato* DC3000 (Schellenberger et al., 2021). In our assays, HAAs had no effect when applied as root treatment to impair nematode establishment. Nematode infection was only reduced in the presence of mRLs and dRLs. However, with the transfer of Arabidopsis onto mRL-supplemented growth medium, plants were heavily stressed and apparently stopped growing. Hence, we assume that the reduced infection observed for this variant is connected to the reduced root mass. In contrast, plant transfer onto dRL-supplemented growth medium did not affect plant growth, but still significantly impaired nematode infection.

Change in sex determination, retarded development of females, and reduced fecundity are typical effects occurring in resistant plants. However, these effects could not be observed. We provide evidence, that particularly dRLs induced plant responses that may contribute to plant defense against the PPN. They could trigger an enhanced oxidative level in roots and boosted ROS synthesis in leaves upon exposure to a second stimulus. Interestingly, plant exposure to RLs was not accompanied by an altered expression of typical defense marker genes in roots.

Regarding the negative impact of mRLs on Arabidopsis, we decided to focus on dRLs to continue with the analysis of the structural relevance of RLs in plant protection against PPNs. First of all, we found that dRLs have no direct nematicidal activity suggesting that the observed infection decrease is mainly linked to the effect of RLs on the plant. Plant reaction towards dRL congeners differed depending on the structure and concentration. We observed an improvement in the efficiency of reducing nematode infection by dRLs with longer acyl chains. Our data indicate that RLs interfere with nematode infection in the early stages of parasitism. Based on our observations together with literature, we suggest two possibilities how RLs affect nematode infection: interference with nematode orientation and/or plant defense priming.

### 4.1. Effect of RLs on nematode parasitism

One key step in early parasitism events is nematode migration towards the host plant rhizosphere. Nematode activity and their attractivity towards the host plant roots are parameters that affect the early parasitism phase. We observed that mRLs and dRLs affect the activity of J2s in a concentration-dependent manner. An increase in concentration led to a decrease in nematode motility. This may give the plant a time

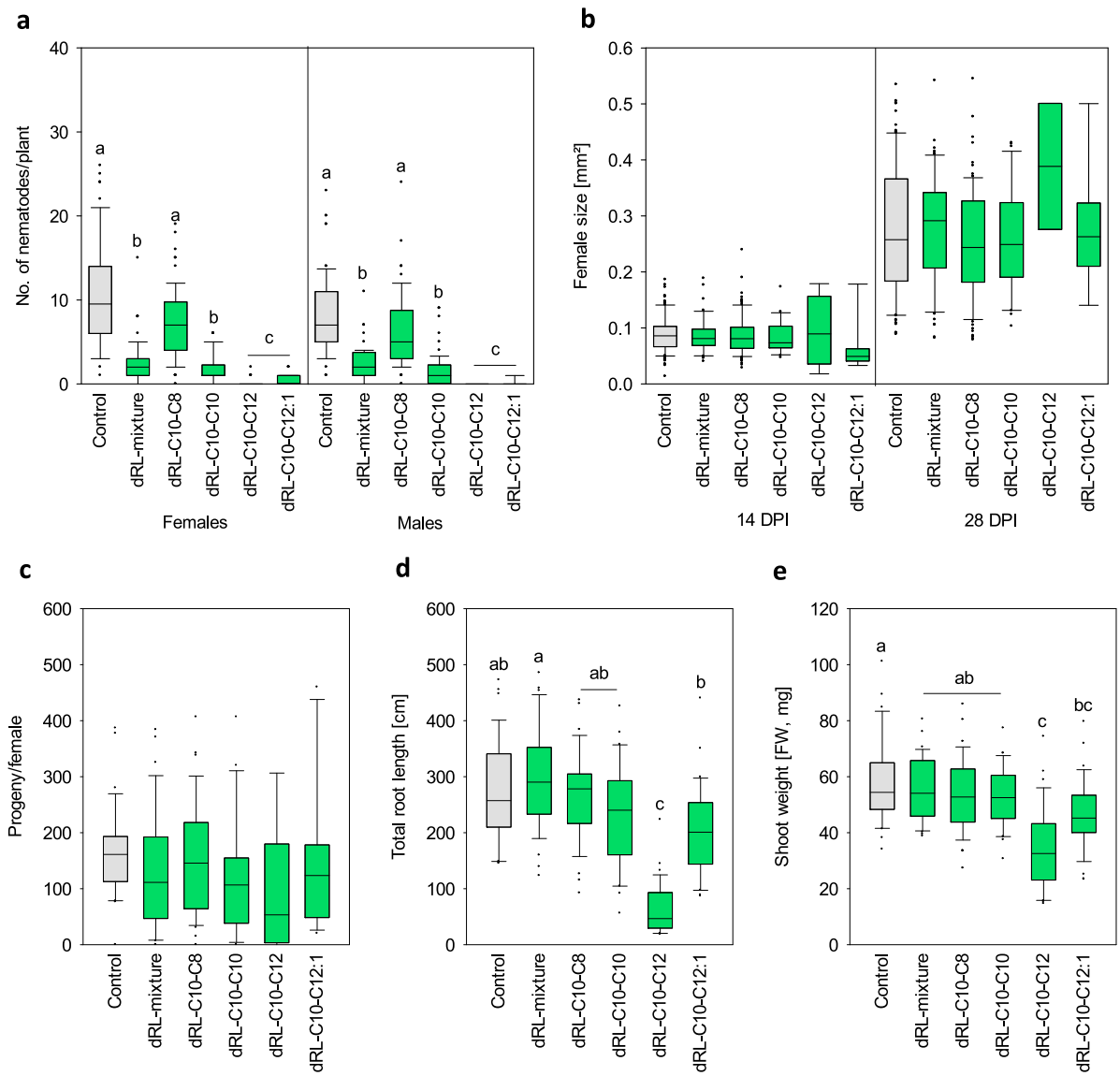


**Fig. 4.** Impact of di-rhamnolipid (dRL) congeners as 8.3 ppm plant growth medium supplement on the infection and plant phenotype of *Arabidopsis thaliana* inoculated with *Heterodera schachtii*. Displayed are (a) the number of adult nematodes at 14 days post inoculation (DPI), (b) the size of females at 14 and 28 DPI, (c) progeny per female at 35 DPI for the infection phenotype, and (d) the total root length (cm), and (e) shoot fresh weight (mg) at 14 DPI for the plant phenotype. Median with 25 %- and 75 %-quartile of three independent biological replicates with total $n \geq 66$ plants for the infection phenotype and total $n \geq 32$ plants for the plant phenotype. Significance based on ANOVA on ranks and post hoc Dunn's Method, different letters indicate significant differences between treatments ($p < 0.05$).

advantage to act early on protection measures since plants are able to sense and react upon nematode-specific molecules (Mendy et al., 2017).

Although neither mRLs nor dRLs show strong direct attraction or repellent activity towards J2s of *H. schachtii*, RLs may interfere with root attractivity. Root exudates play a crucial role to attract nematodes and guide their way to the host plant (Grundler et al., 1991). Zhao et al. (2000) found that J2s of the root-knot nematode *Meloidogyne incognita* did no longer accumulate in the area of the root tip of pea roots when border cells and thus their root exudates were washed off with water before nematode inoculation (Zhao et al., 2000). Considering the tensioactive character of surfactants, RLs are even more potent agents to detach root compounds to alter the characteristic root exudate composition required for nematode orientation.

Interference with nematode orientation could also be linked to a peculiarity of RLs closely resembling ascarosides. These glycolipidic nematode pheromones comprise the dideoxysugar ascarylose linked to a fatty acid-like side chain and play a crucial role in basal nematode activities (Park et al., 2019). Recently, it was observed that *A. thaliana* is able to absorb and process ascarosides turning them into a repellent for PPNs (Manohar et al., 2020). Further studies are required to find out, whether *A. thaliana* is able to also absorb and process bacterial RLs. In 1989, Terzaghi successfully used the synthetic surfactant Tween as conjugate to transfer exogenous fatty acids into different plant species (Terzaghi, 1989). In similar experiments with *A. thaliana*, Tween detergents enabled the uptake of polyhydroxyalkanoates, but the efficiency was dependent on the type of Tween indicating that there are attributes in addition to the general amphiphilic nature that determine the success of this process (Mittendorf et al., 1999). If bacterial RLs can pass the plant membrane like their synthetic counterpart Tween or their nematode-derived structural equivalent ascaroside, a modification of the RL molecules as described for ascarosides may similarly affect nematode orientation outside or inside plants and thus impair nematode parasitism in particular at an early stage.

Another key step during early parasitism is host invasion and the induction of the feeding site. At this stage, host plant defense is highly active (Kammerhofer et al., 2015). However, neither ethylene-, nor

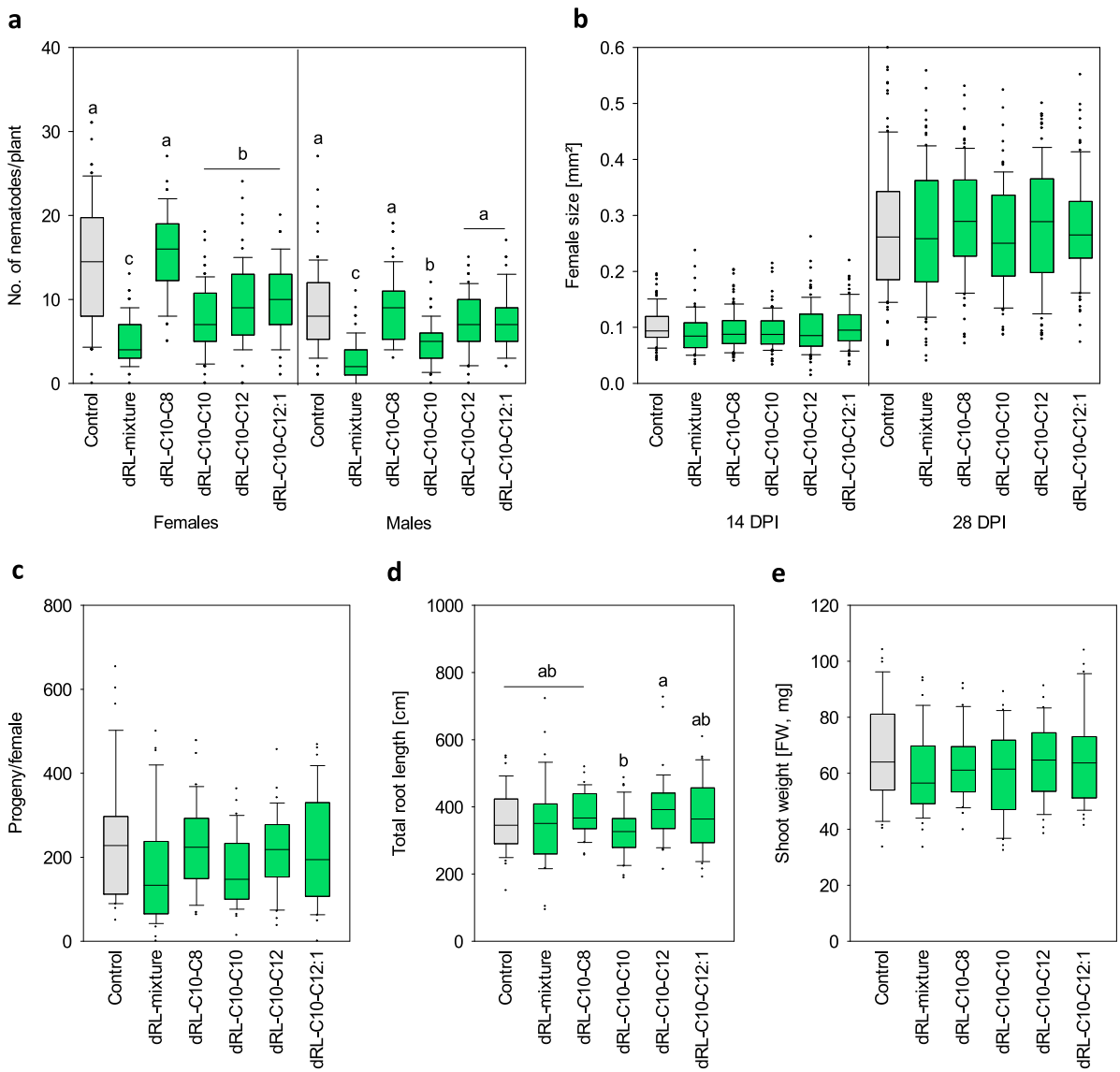


**Fig. 5.** Impact of di-rhamnolipid (dRL) congeners in their relative percentage in a natural dRL mixture at 8.3 ppm as plant growth medium supplement (0.4 ppm dRL-C10-C8, 5.1 ppm dRL-C10-C10, 1.6 ppm dRL-C10-C12, 1,2 ppm dRL-C10-C12:1) on the plant and infection phenotype of *Arabidopsis thaliana* infected with *Heterodera schachtii*. Displayed are (a) the number of adult nematodes at 14 days post inoculation (DPI), (b) the size of females at 14 and 28 DPI, and (c) progeny per female at 35 DPI for the infection phenotype and (d) the total root length (cm) and (e) shoot fresh weight (mg) at 14 DPI for the plant phenotype. Median with 25 %- and 75 %-quartile of three independent biological replicates with total $n \geq 64$ plants for the infection phenotype and total $n \geq 32$ plants for the plant phenotype. Significance based on ANOVA on ranks and post hoc Dunn's Method, different letters indicate significant differences between treatments ($p < 0.05$).

jasmonic acid- or salicylic acid-related defense markers showed altered expression levels in roots in response to 8.3 ppm RLs. Varying observations with different RL mixtures, types and concentrations can be found in literature. Regulation of the tomato defense markers *PR1* and *PRP69* was not affected 3 days after stimulation with 300 μM synthetic mRLs (Robineau et al., 2020). In grapevine cells, 10 ppm of a mixture of 40 % C10-C10 mRLs and 60 % C10-C10 dRLs induced upregulation of the *chit4c* defense marker after 9 h (Varnier et al., 2009). An upregulation of PR1, PR4, and PDF1.2 expression in Arabidopsis leaves was observed after exposure to higher concentrations such as 200 and 1000 ppm RLs. Upregulation was highest between 6 and 24 h and increased with RL concentration (Sanchez et al., 2012). In conclusion, an activation of plant defense genes by RLs is possible, but it is not necessarily the prevalent response. The plant species and the types of tissue (i.e. leaves versus roots) could explain differences in RL-induced immunity, just like the RL type and the used concentration.

### 4.2. RLs as priming agents against PPNs

Defense priming is described as a two-step mechanism initiated by a priming stimulus followed by a stressor as the trigger stimulus (Martinez-Medina et al., 2016). Plant fitness benefits from the primed stage in two ways: upon pathogen attack plant defense activation is more robust while plant maintenance costs are lower in comparison to unprimed defense activation. Correspondingly, in our infection assays we observed that during dRL treatment nematode infection was reduced while root growth was better under conditions of infection pressure.

Effective defense priming activity depends on a functional array of priming and trigger stimulus. Pretreatment with 10 ppm mRLs not only led to an elevated flg22-induced ROS amplitude, but also to an elevated ROS level when applying water instead of flg22 (Fig. 2a). Water is non-stimulative and therefore commonly used as negative control in ROS assays. Since neither the other mRL/water variants nor any of the dRL/water variants showed a difference in total ROS synthesis, we assume

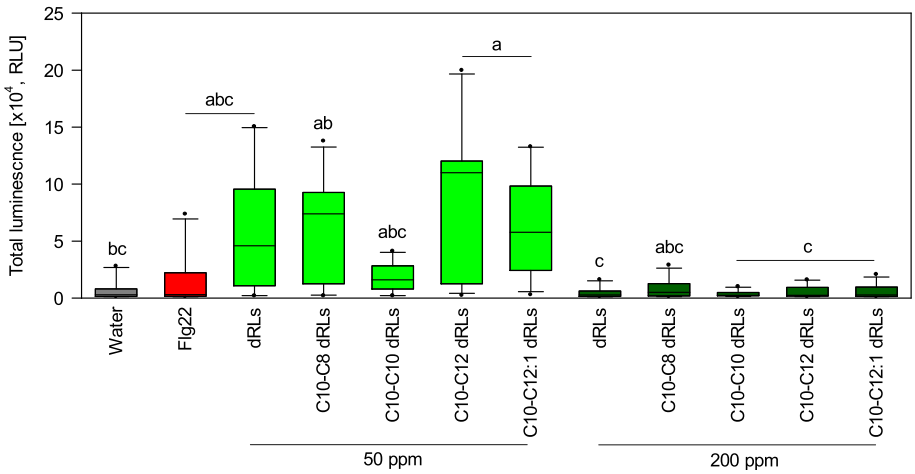


**Fig. 6.** ROS production of *Arabidopsis thaliana* roots in response to 50 ppm and 200 ppm of individual di-rhamnolipid (dRL) congeners and the original dRL mixture monitored over a period of two hours displayed as total ROS production. Median with 25 %- and 75 %-quartile of three independent biological replicates with total $10 \leq n \leq 12$. Significance based on ANOVA on ranks and post hoc Dunn's Method, different letters indicate significant differences between treatments ($p < 0.05$). Relative Light Units (RLU). The corresponding ROS profiles are available in Suppl. Fig. 7.

that a pretreatment with 10 ppm mRLs led to a prolonged ROS burst. Prolonged alterations of the ROS level are known to impair plant development (Morales and Munné-Bosch, 2016; Tsukagoshi, 2016) which may explain the poor phenotype observed for plants grown in the presence of mRLs. However, this also makes it more difficult to interpret the significantly higher total ROS values in response to flg22 after a pretreatment with 10 ppm mRLs. Interestingly, we could not observe an increase in total ROS due to a pretreatment with any dRL concentration, but a change in the flg22-induced curve pattern towards a higher peak for 10 ppm dRL/flg22 (Fig. 2c). Although there was no significant change in the speed of reaction, there is a clear trend for the dRL mixture as well as its individual congeners towards a higher concentration boosting the response in terms of time (Suppl. Fig. 8). A priming character of RLs can also be found in literature. Priming stimulation of grapevine cells with 5 ppm of a mixture of 40 % C10-C10 mRLs and 60 % C10-C10 dRLs led to an increased oxidative burst induced by chitosan, but not by *Botrytis cinerea* culture filtrate (which contains BcPG1 elicitors) (Varnier et al., 2009). In addition, stimulation with 200 ppm of the same RL mixture potentiated the expression of *PR1* in Arabidopsis after inoculation with *B. cinerea* 4 days later, but not when inoculated with *Pseudomonas syringae* pv. *tomato* DC3000 or *Hyaloperonospora arabidopsidis* (Sanchez et al., 2012). Potentiated expression of *PR1* and of *PRP69* in response to *B. cinerea* was also confirmed in tomato after stimulation with 300 μM synthetic mRLs (Robineau et al., 2020).

One adjustment that allow plants to react faster and/or stronger to stressors is the accumulation of PRRs such as the transmembrane receptor kinase FLAGELLIN SENSITIVE2 (FLS2) which regulates the perception of flg22 (Chinchilla et al., 2006; Martinez-Medina et al., 2016). Our data show a boost in ROS synthesis induced by flg22 after RL stimulation, but it is yet unclear if this effect is related to FLS2 accumulation. Future work should further focus on the expression level of PRRs after RL treatment, including PRRs for MAMPs like FLS2 as wells as for nematode-associated molecular patterns (NAMPs). The field of NAMPs is still relatively new, but it was observed that plant reaction to NAMPs, pathogen-associated molecular patterns (PAMPs), and MAMPs is similar suggesting that plants have evolved receptors specific for nematode perception (Manosalva et al., 2015; Mendy et al., 2017). Here, the Arabidopsis leucine-rich repeat receptor-like kinase (NILR1), the first receptor described to be involved in plant immunity to PPNs, would be a suitable candidate (Mendy et al., 2017).

Another adjustment of the plant memory during plant defense priming is histone modification and thus the regulation of transcription factors (Jaskiewicz et al., 2011; Martinez-Medina et al., 2016). However, we could not confirm defense priming activity based on transcription factor expression using MYB72. The root-specific transcription factor MYB72 is described to be essential for defense priming activation induced by root colonization of Arabidopsis with nonpathogenic *Pseudomonas fluorescens* WCS417r, another *Pseudomonas* species described to produce RLs and thus, a potentially suitable candidate for our expression analysis (Van der Ent et al., 2008; Vasileva-Tonkova et al., 2011). More generally, *AtMYB* genes are involved in secondary metabolism which often requires specific conditions to expedite a phenotype (Stracke et al., 2001). The activity of *MYB72* is furthermore associated with iron-deficiency and Palmer et al. suggested that the onset of *MYB72*-related priming may rather be linked to a competition for iron (Palmer et al., 2013). Still, there are further relevant transcription factors, e.g. of the WRKY family, linked to gene priming which should be considered in future experiments to further define priming activity of RLs (Conrath et al., 2015).

Defense priming in plants is confirmed for diverse stimulants, several of them related to bacteria. Bacterial dRLs meet some of the criteria described by Martinez-Medina et al. (2016) including low fitness costs and more robust plant defense. However, the memory effect of the priming stimulus is only indicatively addressed in our experiments. Aspects like accumulation of PRRs, in particular for the perception of PPNs, and the expression of certain transcription factors require more research. Although we cannot yet explain how RLs boost the oxidative burst upon flg22 exposure, we assume RLs prime the leaf immune system for reinforced plant defense upon a second trigger. Perception of the bacterial elicitor flg22 entails an oxidative burst that stimulates plant immune activity (Bauer et al., 2001; Chinchilla et al., 2006). An increase in ROS synthesis as we observed in response to flg22 after RL treatment may reinforce signaling and eventually plant defense. It is possible that similar processes are induced by a RL treatment of plant roots to reinforce plant defense against *H. schachtii*.

### 4.3. RL congeners differ in plant response induction

Pure dRLs demonstrated potential to suppress parasitism of the PPN *H. schachtii*. Interestingly, we saw clear differences in plant response depending on the individual dRL congener applied. Nematode infection was no longer reduced in the presence of C10-C8 dRL while in the presence of dRLs with longer acyl chains infection was reduced again, but the efficiency as well as plant sensitivity strongly differed among dRL congeners and concentrations. The number of female nematodes was as effectively reduced in the presence of dRLs with a C10-C12 or

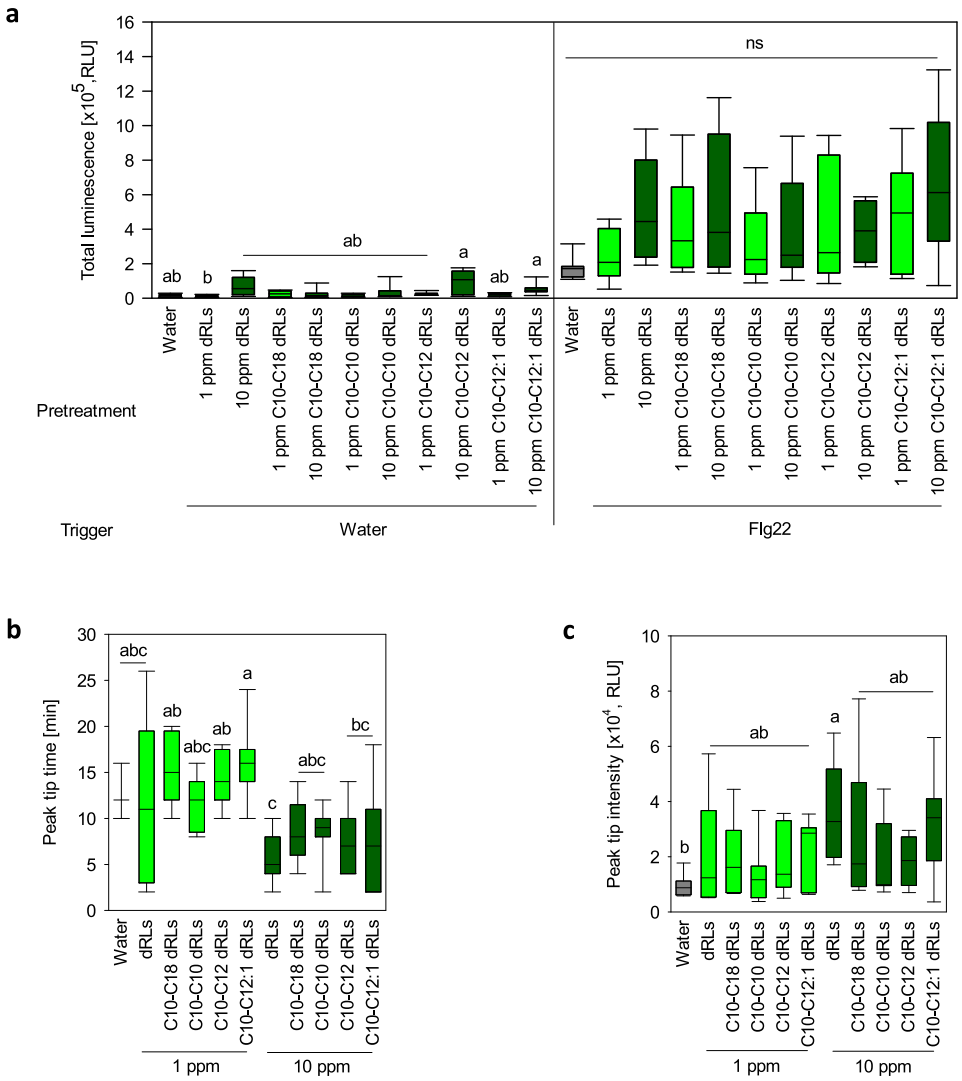


**Fig. 7.** ROS production of *Arabidopsis thaliana* leaf tissue triggered by flagellin 22 (flg22) after pretreatment with different concentrations of di-rhamnolipid (dRL) congeners monitored over a period of two hours. (a) Total ROS produced within the initial 2 h of flg22 induction. (b) Time point the flg22-triggered curve reaches its peak after pretreatment with dRL congeners. (c) Luminescence peak of the flg22-triggered curve after pretreatment with mRLs or dRLs. Median with 25 %- and 75 %-quartile of two independent biological replicates with total $n = 8$. Significance is based on ANOVA on ranks and Tukey test, different letters indicate significant differences between treatments ($p < 0.05$). Relative Light Units (RLU). The corresponding ROS profiles are available in Suppl. Fig. 8.

C10-C12:1 acyl chain at a concentration below 2 ppm as in the presence of C10-C10 dRLs at a concentration above 5 ppm. However, a concentration of 8.3 ppm C10-C12 dRLs caused significant plant growth inhibition for root and shoot tissue while the addition of a single double bond in C10-C12:1 dRLs could avert growth inhibition for shoot tissue. For synthetic RLs with a C12 acyl chain a positive correlation with the biological activity against the fungus *B. cinerea* on tomato and *Zymoseptoria tritici* on wheat was reported (Platel et al., 2021; Robineau et al., 2020).

With 61 %, C10-C10 dRL was the predominant congener in the used mixture. In contrast to the assumption that the most abundant congener determines the overall performance of the mixture, we demonstrate that this congener alone is weaker compared to the dRL mixture in terms of the impact on nematode infection and plant phenotype. Neither the C10-C10 dRL nor any other dRL congener could recreate the priming stimulation of the dRLs mixture. The ROS response directly induced by C10-C10 dRLs was also not similar to that of the dRL mixture, most similar was the one induced by C10-C12 and C10-C12:1 dRLs. The C10-C12:1 dRL was closest by total ROS median and both congeners elicited a small peak in the root ROS profile that was also observed for the mixture. This pattern is particularly interesting because it is often associated with receptor-based recognition (Poncini et al., 2017).

The impact of differences in the acyl chain of RLs as ROS elicitors was shown with tomato leaves and ether-synthetic mRLs with an acyl chain length varying from C4 to C18 inducing either no ROS (C4, C6, C14, C16, and C18), moderate (C8) or strong and long-lasting ROS synthesis (C10 and C12) (Robineau et al., 2020). Interestingly, here the ether-synthetic C10 and C12 mRLs induced a very similar ROS response, but they differed in their approach to control *B. cinerea.* While the ether-synthetic C12 mRLs effectively inhibited mycelium growth, the C10 counterpart was most effective to prevent conidial germination.

Comparing the diverging effects of single congeners we observed on nematode infection, plant development and early plant signaling, we hypothesize that the effect observed for the dRL mixture is the result from a synergism of individual congeners rather than from one dominating congener. The effectivity of RLs most likely changes due to the absence, presence or even minor differences in the concentration of individual RL congeners. Further exploration of the natural resources of RL producers together with genetic engineering and adjusted culture conditions will be a promising approach to further optimize RL compositions towards individual target applications.

In conclusion, the effect of RLs on plant growth and plant infection

with *H. schachtii* differs depending on the molecular structure. Nematode infection was reduced in the presence of both mRLs and dRLs, but pure mRL treatment caused plant growth impairments. The effect of dRLs varied on congener level and revealed high efficiency of dRLs with C10-C12 and C10-C12:1 acyl chains to reduce nematode infection. Here, RLs with longer acyl chains as produced by *Burkholderia* species could be interesting candidates to understand the impact of the acyl chain length of RLs in plant defense stimulation against PPNs. The impact of dRLs on the infection cannot be linked to nematicidal activity, but to the stimulation of plant defense priming. Here, transcriptome sequencing may enable to further elucidate the mode of action of RLs in plant defense against PPNs. In order to validate the potential of RLs in plant protection against PPNs, their impact should be evaluated with crop plants and other important PPN species under non-axenic conditions.

## Funding

This work was supported by the Federal Ministry of Education and Research (BMBF, Germany) as part of the NemaContAnt project [FKZ 031B0352A]. Partial funding was obtained by the Ministry of Innovation, Science, and Research within the framework of the NRW Strategieprojekt Bioeconomy Science Center (No. 313/323–400-002 13) in the FocusLab *Bio2* and the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) under Germany's Excellence Strategy within the Cluster of Excellence FSC 2186 'The Fuel Science Center'. The funders had no influence on the study design, collection, analysis and interpretation of data, and writing the manuscript. Open access funding enabled and organized by Project DEAL.

## CRediT authorship contribution statement

**Sandra Bredenbruch:** Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Conrad Müller:** Writing – original draft, Investigation, Formal analysis. **Henry A. Nvenankeng:** Investigation. **Lukas Schröder:** Investigation. **Antonia C. Zeisel:** Investigation. **Rainier C. Medina:** Investigation. **Till Tiso:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Lars M. Blank:** Writing – review & editing, Resources. **Florian M.W. Grundler:** Writing – review & editing, Supervision, Resources. **A. Sylvia S. Schleker:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

## Declaration of competing interest

This work is part of the patent application "Method of Pathogen Control" with SB, ASSS, FMWG, CM, TT, and LMB being inventors [WO2022238567A1]. The other authors have nothing to declare.

## Data availability

Data will be made available on request.

## Acknowledgments

We thank Ute Schlee and Stefan Neumann for excellent technical and experimental support. The graphical abstract was created with BioRender.com.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pestbp.2024.106103.

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